1. The three elements in question are as different as they are interesting. The only connection between them is that three consecutive elements form an arithmetic series.

The lightest one of them, called “A” from now on, exists in nature only bound to oxygen. As an element it was generated for the first time by Joseph Louis Gay-Lussac und Louis Thénard in 1811. Nowadays “A” is produced by the reduction of its oxide with coke. Purest “A” is gained by the reaction of crude “A” with HCl and subsequent reduction of the respective product with H_2.

Element “A” forms a series of hydrogen compounds, which are analogous to the alcanes. If in the simplest of these compounds two hydrogen atoms are substituted by chlorine and two are replaced by methyl groups, it will be possible after hydrolysis of these substances to synthesize condensation plastics.

The compounds of element “A” with oxygen consist of tetrahedral building blocks, which are placed into the crystal lattice as singles, groups, or chains. The tetrahedrons can also form a three-dimensional frame. A very beautiful deeply blue colored mineral which is also used as gemstones, called lazurite or lapis lazuli, consists of such a three-dimensional frame, in which three of six “A”-atoms are substituted by aluminum atoms. The blue color is a result of S^{3-}-ions. The proportion of the tetrahedrons to the S^{3-}-ions is 6:1. As cations the mineral contains sodium ions. Lapis lazuli always appears together with another very well-known rock. If a finely pulverized sample of lapis lazuli is treated with diluted hydrochloric acid, it will produce a mixture of gases, which on the one hand smells like rotten eggs, and on the other hand clouds a clear solution of calcium hydroxide. Aside from that the formation of colloidal sulfur can be observed.

1.1. Which element is “A”?

1.2. How do you call the hydrogen compounds of “A”?

What is the general formula of these compounds?

1.3. What is the formula and the name of the hydrolyzed chlorine-methyl-compound of the element “A”?

1.4. Write the equation of the polycondensation of the latter. What do call the product of this reaction (umbrella term)?

1.5. The general notation of the molecular formulae for the oxygen compounds usual is $[A_xO_y]_n$. Find depending on x and y a general formula for the charge n of the formula unit in question!

1.6. What is the formula of the oxygen compound of “A”, in which 4 tetrahedrons attached via the corners form a chain, and silver ions act as cations?

1.7. Write down the formula for lapis lazuli?

1.8. What is the accompanying rock of lapis lazuli? Give your reasons by writing a balanced reaction equation!

1.9. Write a balanced ionic equation for the formation of sulphur and the displeasing smelling gas, when powdered lapis lazuli is treated with hydrochloric acid.

1.10. Draw a Lewis formula of the trisulphide ion. Consider the geometry!

The second element "X" received its name from the Greek goddess of the moon and was discovered by Berzelius in 1817. There are three modifications of this element, a black one, a grey one, and a red one. Technically it is extracted from the anode sludge of the copper raffination.

Two oxides of this element are well known, in which the element has a mass fraction of 71.16% or 62.19% respectively. The oxide in which the element has the lower oxidation number, reacts with water to give a weak acid H_aXO_b. The compound formed by the reaction of water with the oxide, where the element has the higher oxidation number, yields the very hygroscopic oxygen acid H_cXO_d.

The standard potential XO_d^{x-}/XO_b^{y-} at pH = 0 amounts to E° = 1.15 V.

The element forms cations with the formulae X_4^{2+}, X_8^{2+} and X_{10}^{2+} respectively, which are stable in complex salts. Additionally, salts are known, where X_2^{2-} ions are present, an example is Na_2X_2.

1.11. Which element is X? Show by a calculation.

1.12. What are the formulae of the two oxygen acids of X?

1.13. Choose one of the two pairs, which may act as oxidizing agents for XO_b^{y-} at pH = 0. Write a balanced ionic equation for this redox reaction.

$E°(Fe^{3+}/Fe^{2+}) = 0.771$ V

$E°(MnO_4^-/Mn^{2+}) = 1.510$ V (for pH = 0)

1.14. The X_4^{2+}-ion is planar. Show that it exhibits aromatic behavior, and draw its Hückel scheme.

Give also its π-binding order.

The third element "Z" is found in nature as a monoatomic gas, and was discovered by Ramsay and his colleagues in 1898. It forms a series of fluorine compounds, ZF_2, ZF_4 and ZF_6. Hydrolysis of the hexafluoride generates a substance which contains only oxygen beside the element, and explodes at 25°C.

1.15. What is the element "Z"?

1.16. Find the oxidation numbers of "Z" in the three given fluorine compounds.

1.17. Give names for the geometry of the structures of the three fluorine compounds of "Z".

1.18. Write a balanced equation for the hydrolysis of ZF_6!

1.19. What is the remarkable theoretical binding behavior in these compounds?

2. HIn is a weakly acidic indicator.

$$HIn + OH^- \rightleftharpoons In^- + H_2O$$

also written as

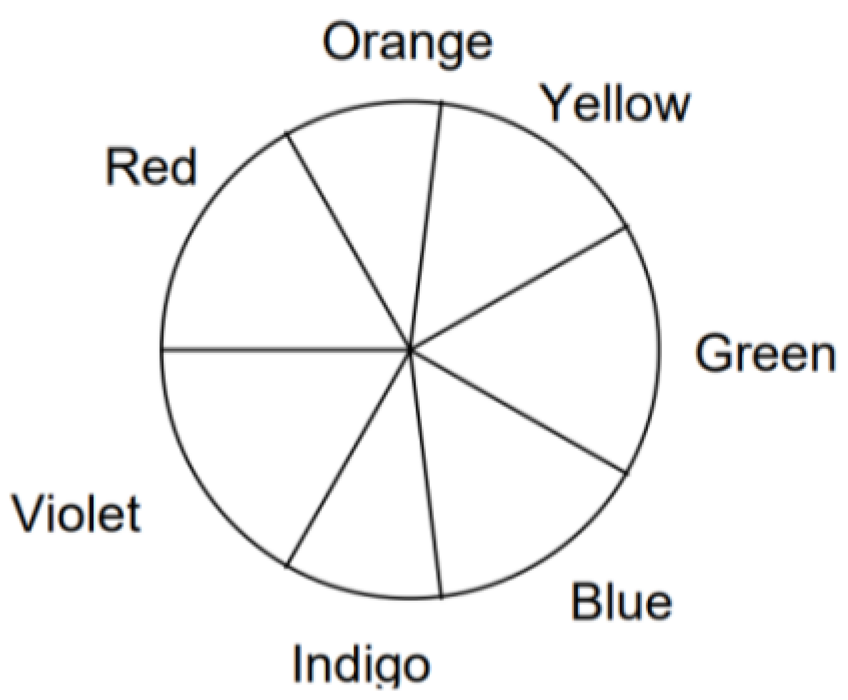
$$HIn \rightleftharpoons In^- + H^+$$

At normal temperatures, the acid dissociation constant for this indicator is $K_a = 2.93 \times 10^{-5}$.

The absorbance data (1.00 cm cells) for 5.00×10^{-4} M solutions of this indicator in strongly acidic and strongly alkaline solutions are given in the following table.

Absorbance Data (A)

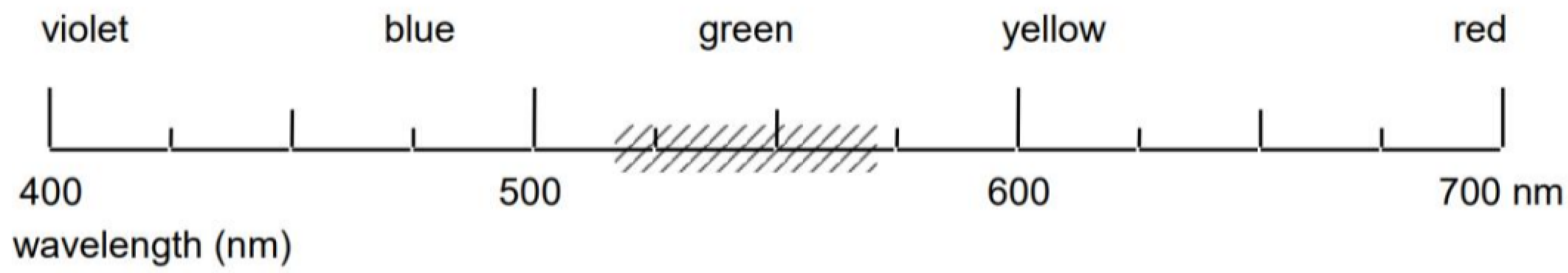
λ, nm	*pH* = 1.00	*pH* = 13.00
400	0.401	0.067
470	0.447	0.050
485	0.453	0.052
490	0.452	0.054
505	0.443	0.073
535	0.390	0.170
555	0.342	0.342
570	0.303	0.515
585	0.263	0.648
615	0.195	0.816
625	0.176	0.823
635	0.170	0.816
650	0.137	0.763
680	0.097	0.588



2.1 Predict the observed color of the a) acidic and b) basic forms of the indicator.

Using a “50 nm wide bar”, shade the appropriate area of the wavelength scale on the answer sheet which would correspond to the color of the indicator at the pH values given in the table.

For example, if observed color is green, your answer would appear as:



2.2 A filter is located between the light source and the sample. What color filter would be most suitable for the photometric analysis of the indicator in a strongly acidic medium?

2.3 What wavelength range would be most suitable for the photometric analysis of the indicator in a strongly basic medium?

2.4 What would be the absorbance of a 1.00×10^{-4} M solution of the indicator in alkaline form if measured at 545 nm in a 2.50 cm cell?

2.5 Solutions of the indicator were prepared in a strongly acidic solution (HCl, pH = 1) and in a strongly basic solution (NaOH, pH = 13). Perfectly linear relationships between absorbance and concentration were observed in both media at 490 nm and 625 nm, respectively.

The molar absorptivities at the two wavelengths are:

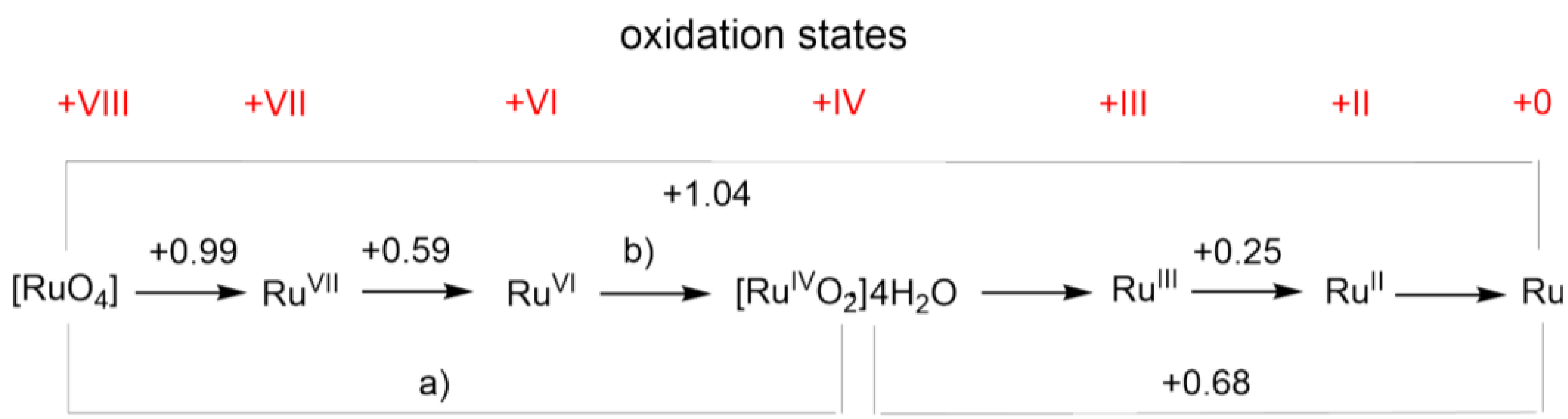
	λ = 490 nm	λ = 625 nm
HIn (HCl)	9.04×10^2 M^{-1} cm^{-1}	3.52×10^2 M^{-1} cm^{-1}
In- (NaOH)	1.08×10^2 M^{-1} cm^{-1}	1.65×10^3 M^{-1} cm^{-1}

Calculate the absorbance (1.00 cm cell) at the two wavelengths for an aqueous 1.80×10^{-3} M solution of the indicator HIn.

3 Two Transition Metals - Many Oxidation States

Among the ruthenium (Ru) isotopes formed during nuclear fission there are two relatively long-lived isotopes, ^{103}Ru and ^{106}Ru, both of which form part of the Highly Active (HA) waste raffinate during spent nuclear fuel reprocessing. Especially the volatile $[RuO_4]$ is a cause for serious concern. In order to investigate the possible mobilization of this metal in nature, the redox properties are intensively studied. The figure below shows the Latimer diagram of ruthenium for acidic conditions.

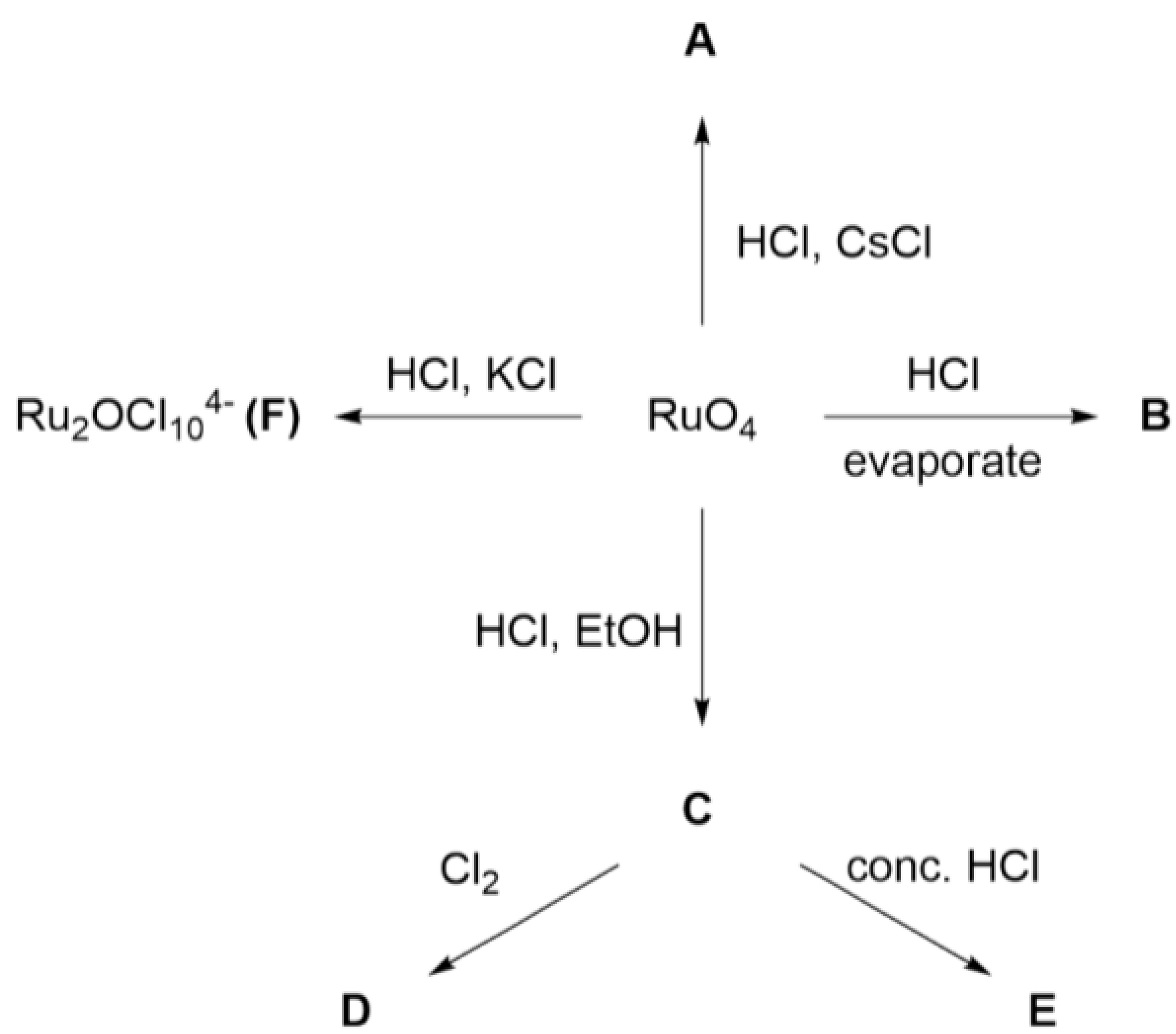
Latimer diagram of ruthenium for acidic conditions (pH 0) vs SHE.



Latimer diagram of Ru for acidic conditions (pH=0) vs SHE (standard hydrogen electrode).

3.1 Calculate the missing potentials a) and b).

Ru forms a number of well-known chloro-complexes. Scheme 2 below shows a choice of the reactions that lead to some of these chloro-complexes starting from [RuO_4].



All compounds **A** to **E** have 6 ligands, either Cl or H_2O or O or a combination of them. Furthermore, the following information about the different species is given:

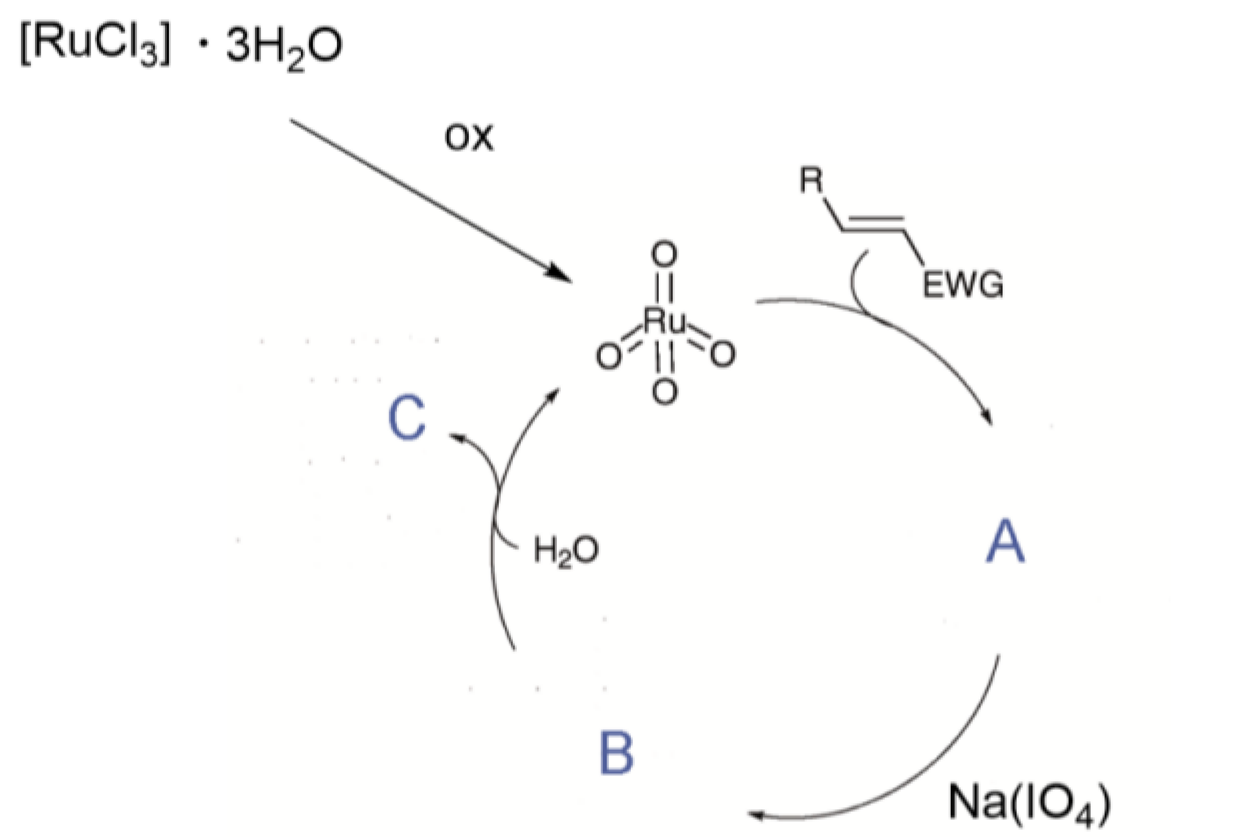
Species	Oxidation State of Ru	Charge of the complex	Molecular Mass (g/mol)
A	+VI	-2	540.69
B	+III		261.48
C	+III	-2	MM > 260
D	+IV	-2	
E	+III	-3	

3.2 Give the formula of compounds **A-E**.

3.3 **F** contains a Ru-O-Ru unit, give the structure of **F**.

3.4 Give the equation for the reaction of [RuO_4] to compound **A**.

Besides the problem for nuclear waste management, ruthenium today is very important for catalysis and bioinorganic chemistry. An interesting example is the application of [$RuCl_3$]·$3H_2O$ in a catalytic reaction called "flash dihydroxylation" (due to its very short reaction time). Mechanistically this reaction is analogous to the very famous Sharpless dihydroxylation with [OsO_4]. Using [$RuCl_3$]·$3H_2O$ and Na(IO_4) as (re)oxidant a large number of syn-dihydroxylation reactions of alkenes have been achieved. The general catalytic cycle of this reaction is depicted in the scheme below.



EWG = electron withdrawing group

3.5 Draw the missing Lewis structures **A**, **B**, and **C**.

3.6 Determine the formal oxidation state of all metal centers.

A similar reactivity (but not catalytic) can be observed by the addition of K[MnO_4] to a basic aqueous solution of an alkene. However, permanganate has a high oxidative potential and can lead to over-oxidation and oxidative cleavage. Table 1 and 2 show the half-reactions of Mn in H_2O at pH 0 and pH 14.

Redox Reactions at pH 0	**E_0 / V**
$Mn^{2+} + 2e^- \rightarrow Mn$	-1.18
$Mn^{3+} + e^- \rightarrow Mn^{2+}$	1.51
$MnO_2 + 4\,H_3O^+ + e^- \rightarrow Mn^{3+} + 6\,H_2O$	0.95
$H_3MnO_4 + H_3O^+ + e^- \rightarrow MnO_2 + 3H_2O$	2.90
$H_2MnO_4 + H_3O^+ + e^- \rightarrow H_3MnO_4 + H_2O$	1.28
$MnO_4^- + 2H_3O^+ + e^- \rightarrow H_2MnO_4 + 2H_2O$	0.92

Table 1

Redox reactions at pH 14	**E_0 / V**
$Mn(OH)_2 + 2e^- \rightarrow Mn + 2OH^-$	-1.56
$Mn_2O_3 + 3H_2O + 2e^- \rightarrow 2Mn(OH)_2 + 2OH^-$	-0.25
$2MnO_2 + H_2O + 2e^- \rightarrow Mn_2O_3 + 2OH^-$	0.15
$MnO_4^{3-} + 2H_2O + e^- \rightarrow MnO_2 + 4OH^-$	0.97
$MnO_4^{2-} + e^- \rightarrow MnO_4^{3-}$	0.27
$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	0.56

Table 2

3.8 At what pH does $[MnO_4^-]$ have the lower reduction potential: pH 0 or pH 14?

3.9 Based on the Frost diagram, are the following species stable? If not write down their reactions in H_2O at pH 0. a) H_3MnO_4 b) Mn^{3+} in H_2O at pH 0?

Overoxidation of an alkene leads to the formation of the corresponding ketone (see example below).

3.10 Draw the missing Lewis structure.

3.11 Give the full redox equation for this example.

4. Nanozymes are nanoparticles (NPs) that exhibit enzymatic activities under different physiological conditions, offering advantages such as high stability, low cost, and tunable activity compared to natural enzymes. Nanozyme $\mathbf{X}_n\mathrm{O}_m$ NPs can be obtained by co-precipitation of salts **S1** and **S2**, both containing element **X**, in alkaline media. 2.7025 g of **S1** and 1.3901 g of **S2** were dissolved in an acidic solvent and treated with tetramethylammonium hydroxide, and 1.1578 g of $\mathbf{X}_n\mathrm{O}_m$ NPs were formed as a result as the only substance containing **X**. The electrospray ionisation mass spectrometry of **S1** revealed positive ion $[\mathbf{X}Hal_2(H_2O)_4]^+$ with m/z 198 and negative ion $[\mathbf{X}Hal_4]^-$ with m/z 196 being the most intense peaks. **S2** lost 45.36% of the total of the original mass and changed colour from green to white above 300 °C. Elemental analysis revealed a mass fraction of 20.09% of element **X** in **S2**.

1. **Find** element **X**, and salts **S1** and **S2**.
2. **Determine** the values of n and m for $\mathbf{X}_n\mathrm{O}_m$ NPs.
3. **Write** the equation of the co-precipitation reaction of $\mathbf{X}_n\mathrm{O}_m$ NPs in ionic form.

Peroxidases are enzymes that catalyse the oxidation of substrates with H_2O_2 as an electron acceptor. Horseradish Peroxidase (HRP) catalyses a typical colourimetric reaction involving H_2O_2 and a chromogenic substrate (H_2TMB_{red}):

$$H_2TMB_{red} + H_2O_2 \xrightarrow{HRP} TMB_{ox} + 2H_2O$$

$\mathbf{X}_n\mathrm{O}_m$ NPs behave like peroxidase due to their ability to participate in the following reaction:

$$\mathbf{X}^{(n-1)+} + H_2O_2 \rightarrow \mathbf{X}^{n+} + OH^* + OH^-$$

4. The peroxidase activity of $\mathbf{X}_n\mathrm{O}_m$ NPs is short-lived. After a certain time, $\mathbf{X}_n\mathrm{O}_m$ NPs are converted into another oxide NPs of element **X** and lose their catalytic activity. **Determine** the chemical formula of the latter, if the mass fraction of **X** in them is smaller than that in $\mathbf{X}_n\mathrm{O}_m$.
5. Which of the following reagents can be used to sustain the catalytic activity of $\mathbf{X}_n\mathrm{O}_m$ NPs? **Choose** the correct answer(s).

   a) $NaIO_4$ b) $NaBH_4$ c) $Na_3C_6H_5O_7$ (sodium citrate) d) Na_3PO_4

In contrast, another $\mathbf{Y}_p$-based nanozyme can act not only as a peroxidase or catalase (converts hydrogen peroxide into water and oxygen) but also can exhibit hydrolase and phosphomonoesterase-like activities without being consumed during the reaction. $\mathbf{Y}_p$ has a spherical structure, in which the **Y** atoms are linked together by covalent bonds. The covalent spherical framework $\mathbf{Y}_p$ has a volume of 1.128×10^{-4} $m^3 \cdot mol^{-1}$ and a surface area of 1.324×10^6 $m^2 \cdot kg^{-1}$.

6. **Identify** element **Y** and number *p*.

The catalytic activity of $\mathbf{Y}_p$-based nanozymes fits Michaelis-Menten kinetics.

The effect of functionalised $\mathbf{Y}_p$-nanozyme was tested with *para*-nitrophenyl phosphate at pH = 7.4 and $T = 37$ °C. The analysis showed $K_M = 1.03$ mM and $r_{max} = 3.73$ $nM \cdot s^{-1}$.

7. The substrate's concentration should be high enough to saturate catalytic sites in nanozymes. If the recorded rate was 0.001 $\mu M \cdot min^{-1}$, **calculate** the corresponding substrate concentration (in μM).

8. If the reaction mixture contained 0.025 mM substrate, **calculate** the time (in min) needed to produce 0.025 μM product.